



Clinical and molecular epidemiology of vancomycin-resistant *Enterococcus faecium* bacteremia from an Indian tertiary hospital

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Abstract

We determined the clinical and molecular epidemiology of emerging nosocomial vancomycin-resistant *Enterococcus faecium* (VREfm)—causing serious bloodstream infections (BSIs) and the correlations between antibiotic resistance and virulence determinants among isolates. All isolates were confirmed by molecular methods (*16S*rRNA and *E. faecium* *ddl* genes) and tested for disk diffusion. PCR was used to detect *aac(6′)-aph(2″)*, *vanA* and *vanB* resistance genes, and *asaI*, *cylA*, *ace*, *esp*, *gelE* and *hyl* virulence genes. VREfm and high-level gentamicin-resistant (HLGR) representative isolates were selected to characterize by pulsed-field gel electrophoresis (PFGE) and multi-locus sequence typing (MLST). Of 173 isolates, 73 (42.2%), 146 (84.4%), and 0 (0.0%) were *vanA*-containing VREfm, *aac(6′)-aph(2″)*-positive HLGR, and *vanB*-positive. Independent predictors of VREfm infection were hematological malignancies ($P = 0.001$) and previous hospitalizations ($P = 0.007$). Observed mortality rate was 34.7%. Independent predictors of BSI-related mortality were endotracheal intubations ($P < 0.001$), gastrointestinal diseases ($P = 0.002$), and pulmonary disease ($P < 0.001$). All VREfm were resistant to vancomycin, teicoplanin, ciprofloxacin, and erythromycin. The *esp*, *hyl*, *ace*, *asaI*, *cylA*, and *gelE* genes were detected at 55.9, 22.5, 2.9, 2.3, 1.7, and 1.2%, respectively. The *esp* gene was significantly associated with VREfm compared to VSEfm ($P = 0.001$). PFGE analysis revealed 23 clones, with 7 major clones. The MLST analysis revealed the following five sequence types: ST80, ST17, ST117, ST132, and ST280, all belonging to CC17. The emergence and expansion of VREfm CC17 with limited antibiotic options in our hospital present a serious public health menace and represent challenges to infection control.

Keywords Vancomycin-resistant *E. faecium* (VREfm) · Bloodstream infections of *E. faecium* · Virulence determinants · Pulsed-field gel electrophoresis (PFGE) · Multi-locus sequence typing (MLST) · Molecular epidemiology

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Introduction

Enterococci are the second leading cause of healthcare-associated infections (HAIs) and health-associated bacteremia, approximately 10% of bacteremias. Today, between 30 and 80% of enterococcal infections are caused by *Enterococcus faecium* [1, 2]. Vancomycin-resistant *Enterococcus faecium* (VREfm) has been disseminated worldwide, particularly in hospital settings [3–5]. In USA, 33% of enterococci and more than 80% of *E. faecium* isolates were resistant to vancomycin and ampicillin [4].

The high plasticity of *E. faecium* genome, the ability to easily acquire novel adaptive traits, and resistance to both vancomycin and ampicillin are approximately 10 times more common in clinical *E. faecium* compared with clinical *E. faecalis*, which provides an advantage to *E. faecium* strains in clinical settings where the antibiotic pressure is high [6, 7].

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In addition, the strains of *E. faecium* are generally equipped with virulence and colonizing factors, such as hyaluronidase (*hyl*) and enterococcal surface protein (*esp*). However, other factors, such as cytolysin (*cylA*), gelatinase (*gelE*), aggregation substance (*asaI*), and collagen-binding protein (*ace*), rarely occur. These virulence determinants may mediate adherence to host tissues, play a role in the formation of biofilms, consistent with their pathogenic role in causing infections [6–8]. To date, a shift in the epidemiology of enterococcal infections has been attributed to the increased ability of a genogroup of *E. faecium* related to human pathogen designated clonal complex 17 (CC17) to colonize the gastrointestinal tract of humans, cause severe diseases [9–11].

To the best of our knowledge, we conducted the first study from India to determine the clinical and molecular characterization of VREfm bacteremia from our hospital by polymerase chain reaction (PCR), pulsed-field gel electrophoresis (PFGE), and multi-locus sequence typing (MLST).

Materials and methods

Study setting and design

This prospective and descriptive study was conducted at the All India Institute of Medical Sciences (AIIMS) tertiary hospital in New Delhi, India. During the study period from January 2010 to June 2014, a total of 173 strains of *E. faecium* were isolated from consecutive, non-duplicated blood samples from hospitalized patients routinely received in the microbiology laboratory. *E. faecium* bacteremia was defined as the isolation of monomicrobial *E. faecium* from one or more aseptically collected blood samples from hospitalized patients during the study period and the associated site of infection in agreement with the patient's condition of bacteremic infection. Single positive blood for *E. faecium* was a true infection when the associated site of infection was compatible with the clinical situation of the bacteremic infection. The Charlson co-morbidity index (CCI) and McCabe score were used to measure co-morbidities [12] and the severity of the underlying disease [13]. This study was approved by the Ethics Committee of the AIIMS, New Delhi, India (Ref. No.: IEST/T-151/2010).

Bacterial identification

The *E. faecium* strains were identified at species level using conventional methods, such as culture, Gram's staining, colony morphology, and biochemical tests, including catalase, bile esculine hydrolysis, growth in 6.5% NaCl, L-pyrrolidonyl-beta-naphthylamide (PYR) test, mannitol test, and arabinose test. These strains were further confirmed by multiplex PCR for the genus level by the *16S rRNA* gene with forward primer 16SrRNA-F: GGATTAGATACCCTGGTAGTCC) and

reverse primer 16SrRNA-R: TCGTTGCGGGACTT AACCCAAC [14] and species level by the D-Ala-D-Ala ligase gene with forward primer FAC1-1(+): GAGTAAATCACTGA ACGA and reverse primer FAC2-1(-): CGCTGATGGTATCG ATTCAT [15]. The optimized primer concentration for each gene (0.2 pmol 16SrRNA, 0.8 pmol ddl-*E. faecium*) was used. The other components used in the PCR were 300 µM deoxyribonucleotide triphosphate (dNTPs), 4 mM MgCl₂, 1× PCR buffer, and 2 U Taq DNA polymerase. The PCR condition for the identification of *E. faecium* was as follows: initial denaturation at 94 °C for 3 min followed by 30 amplification cycles with denaturation at 94 °C for 1 min, annealing at 54 °C for 1 min, and extension at 72 °C for 1 min, and final extension at 72 °C for 7 min. PCR products were analyzed by 2% agarose gel electrophoresis in 0.5× TBE buffer with ethidium bromide at 100 V for 75–90 min. PCR products were visualized under the gel documentation system (Syngene, USA).

Susceptibility testing

Susceptibility testing for ampicillin (10 µg), chloramphenicol (30 µg), ciprofloxacin (5 µg), erythromycin (15 µg) and gentamicin (120 µg) to detect high-level gentamicin resistance (HLGR), linezolid (30 µg), quinupristin/dalfopristin (15 µg), rifampin (5 µg), tetracycline (30 µg), tigecycline (15 µg), teicoplanin (30 µg), and vancomycin (30 µg) was performed by disk diffusion method (Kirby–Bauer) as per the Clinical and Laboratory Standards Institute (CLSI) criteria [16]. The microbroth dilution method was used to test the minimum inhibitory concentration (MIC) of vancomycin and teicoplanin following the CLSI guideline [16]. *Staphylococcus aureus* American Type Culture Collection (ATCC) 25923 and *Enterococcus faecalis* ATCC 29212 strains were used as quality control strains for disk diffusion and dilution methods, respectively [16]. MIC of high-level gentamicin was reconfirmed by E-test.

Detection of antibiotic resistance genes

All *E. faecium* isolates were tested for the presence of vancomycin and high-level gentamycin resistance (HLGR) using primers for *vanA*, *vanB*, and *aac(6')-aph(2'')* genes by multiplex PCR [7, 17].

Detection of virulence determinants

Virulence genes *ace*, *asaI*, *gelE*, *cylA*, *esp* and *hyl* in *E. faecium* strains were tested by multiplex PCR as described previously [7, 8].

Pulsed-field gel electrophoresis (PFGE)

The molecular typing of vancomycin-resistant and HLGR *E. faecium* was characterized by macrorestriction analysis

using *Sma*I by PFGE as described previously [18, 19]. InfoQuest™ FP software v.5.4 (Bio-Rad Laboratories, USA) was used to compare and group PFGE patterns. The PFGE pattern was compared with the unweighted pair-group method with averages (UPGMA) for the generation of dendrogram based on dice coefficients, where the optimization and band position tolerances were set at 1.0 and 1.0%, respectively. A similarity coefficient of $\geq 75\%$ was selected to define the patterns of main PFGE clones, while subtypes of PFGE clones and identical PFGE clones were defined by the cut-off value of 75–99 and 100%, respectively [20].

Multi-locus sequence typing

Representative isolates of different PFGE types of VREfm isolates were selected for MLST analysis. MLST on *E. faecium* isolates was performed as described in the corrected version of Homan et al. which was the changes in *gyd*1 (5'-CAAAGTGGCTTAGCTCCAATGG C-3') and *pur*K2 (5'-TACATAAATCCCGCCTGTTTY-3') primer sequences [21]. MLST data were analyzed using *Enterococcus faecium* MLST databases (<https://pubmlst.org/efaecium/>) [22]. Sequences were handled, edited, formatted, aligned, and compared using the Molecular Evolutionary Genetics Analysis (MEGA) X or Seq Scanner ver. 2.0 and the Online National Center for Biotechnology Information (NCBI) Basic Local Alignment Search Tool (BLAST) program (www.ncbi.nlm.nih.gov/blast).

Statistical analyses

The VREfm-infected patient was considered as “case group,” while the vancomycin-sensitive *Enterococcus faecium* (VSEfm)-infected patient was considered as a “control group.” The odds ratio and its 95% confidence interval (CI) were calculated to examine the association of patient characteristics with isolated *E. faecium*. Multiple logistic regressions were used to identify independent predictors. A chi-squared test was used to compare resistance patterns of *E. faecium* and the correlation between the occurrence of virulence genes and antibiotic resistance. All *P* values were two-tailed, and the *P* value < 0.05 was considered statistically significant. All calculations were carried out using the Stata v.12.1 (StataCorp., USA).

Results

Prevalence and risk factors of VREfm

Of the 173 isolates, 42.2% ($n = 73$) were VREfm and 57.8% ($n = 100$) were vancomycin-susceptible *E. faecium* (VSEfm). The results of a univariate analysis comparing bacteremia due to VREfm and bacteremia due to VSEfm are shown in

Table 1. In a multivariate analysis, independent predictors of VREfm bacteremia compared with VSEfm bacteremia were hematological malignancies (OR 3.88, 95% CI 1.80–8.36, $P = 0.001$) and previous hospitalization (< 6 months) (OR 2.72, 95% CI 1.32–5.59, $P = 0.007$).

The observed mortality rate was 34.7%. Factors associated with mortality in patients with *E. faecium* bacteremia are presented in Table 2. In a multivariate analysis, independent predictors of mortality due to *E. faecium* bacteremia were rapidly fatal status according to McCabe score at the time of admission (OR 2.38, 95% CI 1.54–3.60, $P < 0.001$), endotracheal intubations (OR 5.87, 95% CI 2.60–13.26, $P < 0.001$), gastrointestinal diseases (OR 3.68, 95% CI 1.61–8.46, $P = 0.002$), and pulmonary disease (OR 10.67, 95% CI 2.86–39.73, $P < 0.001$).

Antimicrobial resistance and virulence factors

The antimicrobial susceptibility patterns of 100 VSEfm and 73 VREfm isolates are summarized in Table 3. The strains of VREfm were significantly more multidrug-resistant (MDR) than the strains of VSEfm ($P < 0.001$). The PCR results showed that 42.2% of the isolates were *vanA* positive with resistance to both vancomycin and teicoplanin by the MIC test, while all isolates were *vanB* negative. A total of 84.4% of the isolates were HLGR by *aac*(6)-*aph*(2") gene. A total of 95.9% of the isolates from VREfm and 77.0% of the isolates from VSEfm were HLGR. All *vanA*-containing VREfm were resistant to vancomycin, teicoplanin, ciprofloxacin, and erythromycin. More than 95% of the VRE isolates were resistant to ampicillin, gentamicin, rifampicin, and quinopristine-dalfopristine. However, 27.4 and 52.1% of VREfm isolates were chloramphenicol and tetracycline-resistant, respectively. All strains of VREfm ($n = 73$, 100.0%) were MDR, while 87% ($n = 87.0\%$) of VSEfm strains were MDR ($P = 0.001$). All strains were uniformly susceptible to linezolid and tigecycline.

The *esp*, *hyl*, *ace*, *asa*1, *cylA*, and *gelE* genes were detected at 54.9% ($n = 95$), 22.5% ($n = 39$), 2.9% ($n = 5$), 2.3% ($n = 4$), 1.7% ($n = 3$), and 1.2% ($n = 2$) in all *E. faecium* strains ($n = 173$), respectively. Percentages of the *esp*, *hyl*, *ace*, *asa*1, *cylA*, and *gelE* genes in the VREfm strains ($n = 73$) were 69.9% ($n = 51$), 26.0% ($n = 19$), 5.5% ($n = 4$), 4.1% ($n = 3$), 1.4% ($n = 1$), and 1.4% ($n = 1$), respectively. However, the percentages of *esp*, *hyl*, *ace*, *asa*1, *cylA*, and *gelE* genes in the strains of VSEfm ($n = 100$) were 44.0% ($n = 44$), 20.0% ($n = 20$), 1.0% ($n = 1$), 1.0% ($n = 1$), 2.0% ($n = 2$), and 1.0% ($n = 1$). The *esp* gene was significantly associated with VREfm (69.9%) compared with VSEfm (44.0%) ($P = 0.001$).

The association between the presence of *esp* and *hyl* genes and antibiotic resistance between the strains of VREfm and VSEfm is shown in Table 4. Among all *esp*-positive isolates, VREfm strains were significantly associated with most

Table 1. Comparison of demographics, clinical characteristics, and outcomes of patients with vancomycin-resistant (VR) and vancomycin-susceptible *E. faecium* (VSEfm) bacteremia ($n = 173$)

Characteristics	VSEfm bacteremia, $n = 100$	VREfm bacteremia, $n = 73$	OR	95% CI	<i>P</i>
Age, mean $\pm$ SD	41.11 $\pm$ 23.27	35.06 $\pm$ 23.48	—	—	0.09
Sex					
Male	62 (62.0)	42 (57.5)	1.00		0.55
Female	38 (38.0)	31 (42.5)	1.20	0.65–2.23	
Hospitalization ward					
Medical	35 (35.0)	43 (58.9)	1.00		
Surgical	24 (24.0)	9 (12.3)	0.30	1.26–0.74	0.009
ICU	41 (41.0)	21 (28.8)	0.42	0.21–0.83	0.01
LOS before culture (days), mean $\pm$ SD	14.48 $\pm$ 16.64	22.97 $\pm$ 42.16	1.01	1.00–1.03	0.11
0–10	57 (57.0)	31 (42.5)	1.00		
11–20	22 (22.0)	15 (20.5)	1.25	0.57–2.76	0.57
> 20	21 (21.0)	27 (37.0)	2.36	1.15–4.85	0.02
McCabe score, mean $\pm$ SD	1.37 $\pm$ 0.97	1.82 $\pm$ 1.17	1.50	1.11–2.01	0.007
Charlson comorbidity index, mean $\pm$ SD	1.43 $\pm$ 1.10	2.04 $\pm$ 1.50	1.46	1.13–1.89	0.004
Risk factors associated					
Previous hospitalization (< 6 months)	18 (18.0)	30 (41.1)	3.18	1.59–6.34	0.001
ICU admission before culture	59 (59.0)	37 (50.7)	0.71	0.39–1.31	0.28
Inter-hospital transfer	4 (4.0)	5 (6.8)	1.77	0.46–6.81	0.41
Intra-hospital transfer	72 (72.0)	52 (71.2)	0.96	0.49–1.88	0.91
Intravenous catheter in use at time of culture	94 (94.0)	71 (97.3)	2.27	0.44–11.56	0.33
Urinary catheter in use at time of culture	60 (60.)	34 (46.6)	0.58	0.32–1.07	0.08
Prosthetic devices	1 (1.0)	0 (0.0)	—	—	0.39
Endotracheal intubations	33 (33.0)	20 (27.4)	0.77	0.40–1.49	0.43
Underlying diseases/conditions					
Hematological malignancies	13 (13.0)	44 (60.3)	4.41	2.09–9.32	< 0.001
Solid tumor	17 (17.0)	9 (12.3)	0.69	0.29–1.64	0.4
Liver disease	16 (16.0)	4 (5.5)	0.30	0.10–0.95	0.04
Gastrointestinal disease	11 (11.0)	4 (5.5)	0.47	0.14–1.54	0.21
Diabetes mellitus	17 (17.0)	10 (13.7)	0.78	0.33–1.81	0.78
Renal disease	5 (5.0)	6 (8.2)	1.70	0.50–5.81	0.40
Endocarditis	2 (2.0)	1 (1.4)	0.68	0.061–7.65	0.76
Neurological disease	22 (22.0)	7 (9.6)	0.38	0.15–0.94	0.03
Genito-urinary tract disease	3 (3.0)	2 (2.7)	0.91	0.15–5.59	0.92
Immune deficiency	29 (29.0)	32 (43.8)	1.91	1.02–360	0.04
Sepsis	20 (20.0)	15 (20.5)	1.03	0.49–2.19	0.93
Corticosteroids	20 (20.0)	16 (21.9)	1.12	0.54–2.35	0.76
Surgical intervention	39 (39.0)	20 (27.4)	0.59	0.31–1.13	0.11
Recent antimicrobial therapy					
Penicillins ^a	35 (35.0)	30 (41.1)	1.30	0.70–2.41	0.41
Aminoglycosides	44 (44.0)	33 (45.2)	1.05	0.57–1.93	0.88
Metronidazole	26 (26.0)	23 (31.5)	1.31	0.67–2.55	0.43
Fluoroquinolones	40 (40.0)	24 (33.3)	0.75	0.40–1.41	0.37
Linezolid	6 (6.0)	20 (27.4)	5.91	2.24–15.63	< 0.001
Glycopeptides	35 (35.0)	32 (43.8)	1.45	0.78–2.69	0.24
Clindamycin	7 (7.0)	6 (8.2)	1.19	0.38–3.70	0.76
Carbapenem	30 (30.0)	26 (35.6)	1.29	0.68–2.45	0.44
Cephalosporin (3rd generations)	78 (78.0)	48 (65.8)	0.54	0.28–1.07	0.076

Table 1. (continued)

Characteristics	VSEfm bacteremia, <i>n</i> = 100	VREfm bacteremia, <i>n</i> = 73	OR	95% CI	<i>P</i>
Total number of antibiotics received, mean $\pm$ SD	3.75 $\pm$ 2.18	4.12 $\pm$ 4.12	1.07	0.94–1.22	0.29
Monotherapy	7 (7.0)	3 (4.1)	1.00		
Dual combination therapy	22 (22.0)	17 (23.3)	1.80	0.40–8.03	0.44
Triple or multiple combination therapy	67 (67.0)	51 (69.9)	1.78	0.44–7.21	0.42
No antibiotic therapy	4 (4.0)	2 (2.7)	1.17	0.13–10.22	0.89
Outcome					
Mortality	33 (33.0)	27 (37.0)	1.19	0.63–2.24	0.59
LOS total (days), mean $\pm$ S.D	26.25 $\pm$ 21.90	34.47 $\pm$ 43.31	1.01	1.00–1.02	0.14
LOS (days) after culture, mean $\pm$ S.D	11.33 $\pm$ 13.00	11.48 $\pm$ 13.78	1.00	0.98–1.02	0.94
LOS (days) after culture death excluded, mean $\pm$ S.D	13.76 $\pm$ 14.67	13.87 $\pm$ 14.69	1.00	0.98–1.03	0.97
LOS (days) after culture death excluded (> 6 days), mean $\pm$ S.D	20.65 $\pm$ 15.54	19.23 $\pm$ 15.74	0.99	0.96–1.03	0.7

OR, odds ratio; CI, confidence interval; ICU, intensive care unit; LOS, length of hospital stay

Data are no. (%) of patients, unless otherwise specified

P values ≤ 0.05 are highlighted in bold

^a Penicillins include β -lactams– β -lactamase inhibitor combinations

antibiotics compared with VSEfm strains ($P = 0.024$ for ampicillin; $P = 0.047$ for high-level gentamicin; $P = 0.002$ for chloramphenicol; $P = 0.028$ for erythromycin; $P = 0.012$ for rifampicin; $P = 0.001$ for tetracycline; $P = 0.024$ for quinopristin-dalfopristin). The *esp*-positive VREfm strains were significantly associated with MDR compared with *esp*-positive VSEfm strains ($P = 0.028$). However, *hyl*-positive VREfm strains were only significantly associated with high-level gentamicin compared with *hyl*-positive VSEfm strains ($P = 0.004$) (Table 4).

Analysis of clonal relatedness by PFGE and MLST

Representative isolates from VREfm strains (*vanA* positive), *aac(6')-aph(2'')*-positive HLGR isolates, VSEfm isolates, different *E. faecium* antimicrobial resistance profiles, and virulence genes, such as *esp*- and *hyl*-positive isolates, were selected for PFGE typing. PFGE typing was performed on 88 *E. faecium* isolates, of which 3 strains (3.4%) were vancomycin resistant only, 13 (14.8%) were HLGR only, 65 strains (73.9%) were resistant to both vancomycin and high-level gentamicin, and 7 (7.9%) strains were susceptible to vancomycin and high-level gentamicin. The *SmaI* macro-restriction fragment profiles revealed 23 clones named from 1 to 23, with 7 major clusters predominating (clone 1, $n = 13$, 14.8%; clone 9, $n = 12$, 13.6%; clone 7, $n = 11$, 12.5%; clone 6, $n = 7$, 7.8%; clone 10, $n = 5$, 5.7%) and together representing 54.5% ($n = 48$) of *E. faecium* selected for PFGE. Each major clone consisted of 4–11 subtypes. Clone 1 isolates carried *vanA* (100.0%) and *aac(6')-aph(2'')* (84.6%) genes, often containing *esp* gene (84.6%) and *hyl* gene (15.4%). Clone 9 isolates carried *vanA* (83.3%) and *aac(6')-aph(2'')* (100.0%) genes and

often containing *esp* gene (8.3%) and *hyl* gene (33.3%). Clone 7 isolates carried *vanA* (81.8%) and *aac(6')-aph(2'')* (100.0%) genes and often containing *esp* gene (54.5%) and *hyl* gene (18.2%) (Fig. 1).

The MLST typing was performed on a subset of 29 VREfm based on the PFGE patterns of the *E. faecium* isolates. There are five different sequence types (STs)—ST80, $n = 19$; ST17, $n = 7$; ST117, $n = 1$; ST132, $n = 1$; and ST280, $n = 1$, identified, and all of them belong to CC17. A single locus difference between ST17 and ST117, ST132, and ST280 was observed, while a double locus difference was found between ST17 and ST80. Nucleotide sequences were deposited in the GenBank with GenBank accession numbers: MT746109 to MT746129, which originated from the seven housekeeping genes for the MLST of *E. faecium* isolates. The association of antibiotic resistance, virulence genes, PFGE types, and ST of *E. faecium* is shown in Table 5 and Fig. 1.

All ST80 ($n = 19$) isolates were resistant to ampicillin, ciprofloxacin, erythromycin, teicoplanin, and vancomycin, while 94.7% ($n = 18$) of ST80 isolates were resistant to gentamicin and rifampicin and 42.1% ($n = 8$) of ST80 isolates were resistant to tetracycline. Both *esp* and *hyl* genes were detected in 73.7% ($n = 14$) and 42.1% ($n = 8$) of the isolates, respectively. Both *esp* and *hyl* genes were associated with 36.8% ($n = 7$) of the isolates. Both *asaI* and *ace* genes were detected only in 10.5 and 5.3% of the isolates, respectively. These ST80 strains were recovered from hospitalized patients in the medical wards (52.6%, $n = 10$), ICUs (26.3%, $n = 5$), and surgical wards (21.1%, $n = 4$).

All ST17 ($n = 7$) isolates were resistant to ampicillin, ciprofloxacin, erythromycin, gentamicin, and rifampicin, while 85.7% ($n = 6$) of ST17 isolates were resistant to teicoplanin

Table 2. Factors associated with in-hospital mortality among patients with *E. faecium* bacteremia

Characteristics	Patients who survived, n = 113	Patients who died, n = 60	OR	95% CI	P
Age, mean $\pm$ SD	35.55 $\pm$ 22.92	44.21 $\pm$ 23.67	1.02	1.00–1.03	0.002
Sex					
Male	71 (62.8)	33 (55.0)	1		0.32
Female	42 (37.2)	27 (45.0)	0.73	0.38–1.37	
Hospitalization ward					
Medical	55 (48.7)	23 (38.3)	1		
Surgical	26 (23.0)	7 (11.7)	0.64	0.24–1.69	0.37
ICU	32 (28.3)	30 (50.0)	2.24	1.12–4.50	0.02
LOS before culture (days), mean $\pm$ SD	17.63 $\pm$ 34.16	18.88 $\pm$ 21.66	1.00	0.99–1.01	0.80
McCabe score, mean $\pm$ S.D	1.40 $\pm$ 1.10	1.87 $\pm$ 0.98	1.52	1.12–2.06	0.008
Charlson comorbidity index, mean $\pm$ SD	1.50 $\pm$ 0.96	2.03 $\pm$ 1.77	1.36	1.06–1.74	0.015
Risk factors associated					
Previous hospitalization (< 6 months)	34 (30.1)	14 (23.3)	0.71	0.34–1.45	0.34
ICU admission before culture	55 (48.7)	41 (68.3)	2.28	1.18–4.39	0.014
Intravenous catheter in use at time of culture	108 (95.6)	57 (95.0)	0.88	0.20–3.81	0.86
Urinary catheter in use at time of culture	54 (47.8)	40 (66.7)	2.19	1.14–4.19	0.019
Endotracheal intubations	24 (21.2)	29 (48.3)	3.47	1.76–6.83	< 0.001
Underlying diseases/conditions					
Hematological malignancies	27 (23.9)	15 (25.0)	1.06	0.51–2.20	0.87
Solid tumor	19 (16.8)	10 (16.7)	0.99	0.43–2.29	0.98
Gastrointestinal disease	24 (21.2)	24 (40.0)	2.47	1.25–4.91	0.01
Diabetes mellitus	13 (11.5)	14 (23.3)	2.34	1.02–5.38	0.04
Pulmonary disease/asthma	8 (7.1)	10 (16.7)	2.63	0.98–7.06	0.05
Renal disease	7 (6.2)	4 (6.7)	1.08	0.30–3.85	0.90
Genito-urinary tract disease	3 (2.7)	2 (3.3)	1.26	0.21–7.78	0.80
Immune deficiency	42 (37.2)	19 (31.7)	0.78	0.40–1.52	0.47
Sepsis	20 (17.7)	15 (25.0)	1.55	0.73–3.31	0.26
Corticosteroids	24 (21.2)	12 (20.0)	0.93	0.43–2.02	0.85
Surgical intervention	35 (31.0)	24 (40.0)	1.49	0.77–2.85	0.23
Vancomycin resistant <i>E. faecium</i>	46 (40.7)	27 (45.0)	1.92	0.63–2.24	0.59
Total number of antibiotics received, mean $\pm$ SD	3.58 $\pm$ 2.20	4.52 $\pm$ 2.38	1.19	1.04–1.37	0.013
LOS total (days), mean $\pm$ SD	31.52 $\pm$ 36.70	26.32 $\pm$ 23.83	0.99	0.98–1.01	0.33
LOS (days) after culture, mean $\pm$ SD	13.81 $\pm$ 14.61	6.69 $\pm$ 8.54	0.92	0.89–0.97	0.001

OR, odds ratio; CI, confidence interval; ICU, intensive care unit; LOS, length of hospital stay

Data are no. (%) of patients, unless otherwise specified

P values $\leq$ 0.05 are highlighted in bold

Table 3. Comparison of antimicrobial resistance patterns of VS *E. faecium* (VSEfm) and VR *E. faecium* (VREfm) causing bacteremia (n = 173)

Antibiotics	VSEfm bacteremia, n = 100	VREfm bacteremia, n = 73	P
Ampicillin	78 (78.0)	70 (95.9)	0.001
Gentamicin	76 (76.0)	70 (95.9)	< 0.001
Chloramphenicol	5 (5.0)	20 (27.4)	< 0.001
Ciprofloxacin	90 (90.0)	73 (100.0)	0.005
Erythromycin	90 (90.0)	73 (100.0)	0.005
Rifampicin	77 (77.0)	70 (95.0)	0.001
Tetracycline	33 (33.0)	38 (52.1)	0.012
Quinopristin–dalfopristin	92 (92.0)	72 (98.6)	0.05

All P values determined using χ^2 test. P values $\leq$ 0.05 are highlighted in bold

All VREfm were resistant to both vancomycin and teicoplanin

All isolates were susceptible to linezolid and tigecycline

Table 4. Association between presence of virulence determinants (*esp* and *hyl* genes) and antimicrobial resistance of VREfm (*n* = 73) and VSEfm (*n* = 100) causing bacteremia

Antibiotics	VREfm (<i>n</i> = 73)			VSEfm (<i>n</i> = 100)			VREfm (<i>n</i> = 73)			VSEfm (<i>n</i> = 100)			Pf
	<i>esp</i> negative (<i>n</i> = 22)	<i>esp</i> positive (<i>n</i> = 51)	<i>Pa</i>	<i>esp</i> negative (<i>n</i> = 56)	<i>esp</i> positive (<i>n</i> = 44)	<i>Pb</i>	<i>hyl</i> negative (<i>n</i> = 54)	<i>hyl</i> positive (<i>n</i> = 19)	<i>Pd</i>	<i>hyl</i> negative (<i>n</i> = 80)	<i>hyl</i> positive (<i>n</i> = 20)	<i>Pe</i>	
Ampicillin	21 (95.5)	49 (96.1)	0.902	42 (75.0)	36 (81.8)	0.414	52 (96.3)	18 (94.7)	0.768	61 (73.3)	17 (85.0)	0.398	0.316
Gentamicin	21 (95.5)	49 (96.1)	0.902	39 (69.6)	37 (84.1)	0.093	51 (94.4)	19 (100.0)	0.294	63 (78.8)	13 (65.0)	0.198	0.004
Chloramphenicol	5 (22.7)	15 (29.4)	0.557	3 (5.4)	2 (4.5)	0.853	16 (29.6)	4 (21.1)	0.471	4 (5.0)	1 (5.0)	1.000	0.134
Ciprofloxacin	22 (100.0)	51 (100.0)	–	49 (87.5)	41 (93.2)	0.347	54 (100.0)	19 (100.0)	–	70 (87.5)	20 (100.0)	0.096	–
Erythromycin	22 (100.0)	51 (100.0)	–	50 (89.3)	40 (90.9)	0.788	54 (100.0)	19 (100.0)	–	70 (87.5)	20 (100.0)	0.096	–
Rifampicin	21 (95.5)	49 (96.1)	0.902	42 (75.0)	35 (79.5)	0.592	54 (100.0)	16 (84.2)	0.003	60 (75.0)	17 (85.0)	0.342	0.946
Tetracycline	8 (36.4)	30 (58.8)	0.078	22 (39.3)	1 (25.0)	0.132	29 (53.7)	9 (47.4)	0.634	24 (30.0)	9 (45.0)	0.202	0.882
Quinopristin–dalbopristin	21 (95.5)	51 (100.0)	0.125	52 (92.9)	40 (90.9)	0.722	53 (98.1)	19 (100.0)	0.550	72 (90.0)	20 (100.0)	0.140	–
MDR	22 (100.0)	51 (100.0)	–	47 (83.9)	40 (90.9)	0.303	54 (100.0)	19 (100.0)	–	68 (85.0)	19 (95.0)	0.323	–

All *P* values determined using χ^2 test. *P* values ≤ 0.05 are highlighted in bold

^a *P* values derived from *esp*-negative vs. *esp*-positive strains of VREfm

^b *P* values derived from *esp*-negative vs. *esp*-positive strains of VSEfm

^c *P* values derived from *esp*-positive strains of VREfm vs. *esp*-positive strains of VSEfm

^d *P* values derived from *hyl*-negative vs. *hyl*-positive strains of VREfm

^e *P* values derived from *hyl*-negative vs. *hyl*-positive strains of VSEfm

^f *P* values derived from *hyl*-positive strains of VREfm vs. *hyl*-positive strains of VSEfm

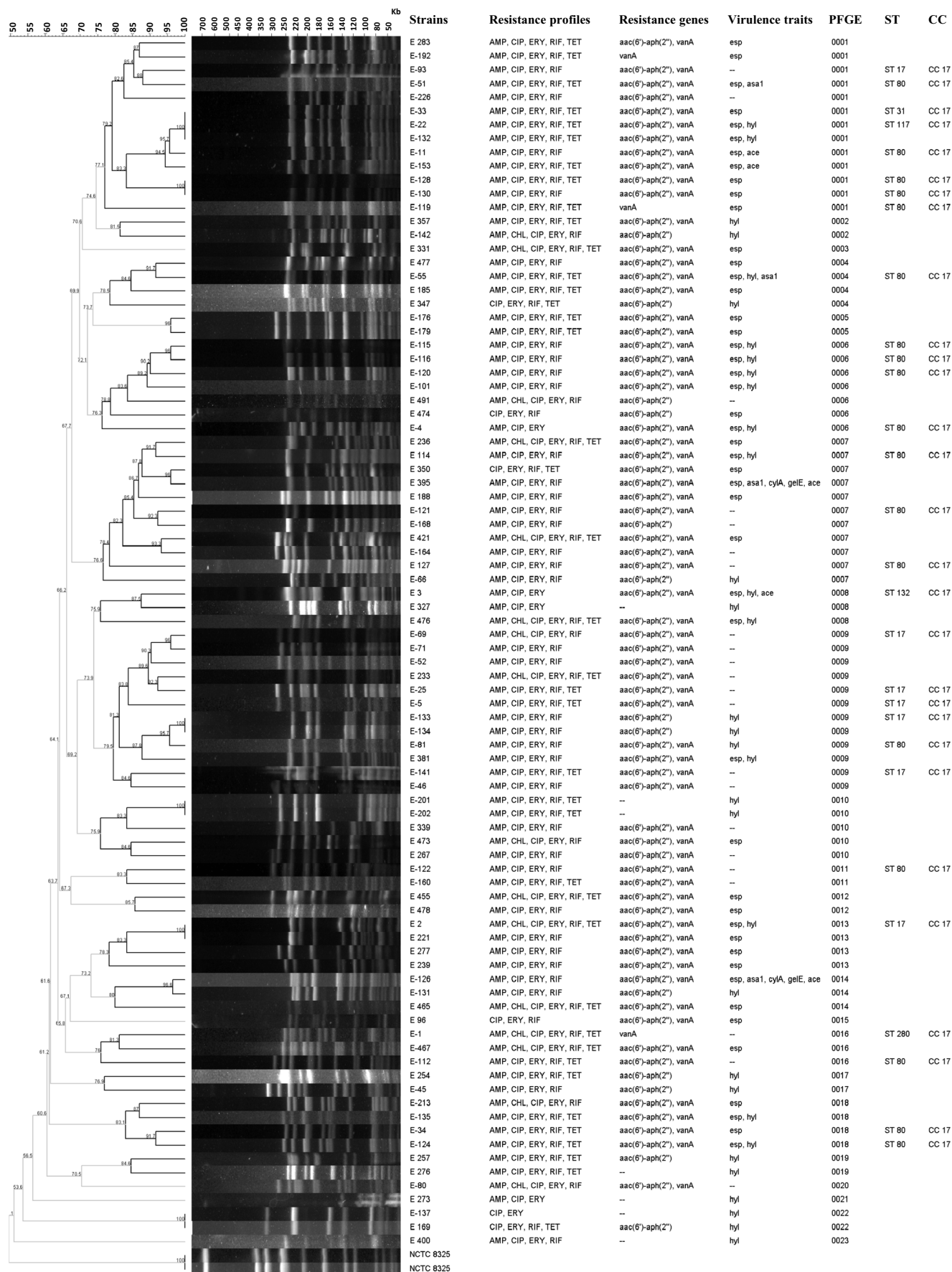


Fig. 1 Clonal relatedness of 88 *E. faecium* isolates as based on *Sma*I macrorestriction analysis in PFGE and subsequent phylogenetic analysis (Dice coefficient using UPGMA clustering; InfoQuest™ FP Software v. 5-4, Bio-Rad Laboratories, USA; settings: optimization 1.0%; tolerance 1.0%). PFGE patterns of *E. faecium* isolates were delineated with a 75% similarity cut-off value [20] and indicated by dark lines. Legend: AMP, ampicillin; CHL, chloramphenicol; CIP, ciprofloxacin; ERY, erythromycin; RIF, rifampicin; TET, tetracycline; *aac(6')-aph(2'')*. PCR results for *aac(6')-aph(2'')* genes; *esp*, *hyl*, *asa1*, *cylA*, *gelE*, and *ace*. PCR results for *esp* or *hyl* or *asa1* or *cylA* or *gelE* or *ace* genes. PFGE, PFGE types; ST, MLST sequence type; CC, clonal complex

and vancomycin. Resistance to tetracycline and chloramphenicol was 57.1% ($n = 4$) and 28.8% ($n = 2$), respectively. These strains were positive for *hyl* (28.8%, $n = 2$). These ST17 strains recovered from hospitalized patients in the medical wards (71.4%, $n = 5$), ICUs (14.3%, $n = 1$), and surgical wards (14.3%, $n = 1$) (Table 5).

Discussion

This work is the first prospective study on *E. faecium* molecular epidemiology isolated from blood samples in an Indian hospital. In this study, we reported a higher prevalence of VREfm than previously reported [23, 24]. Similar to our study, VREfm was reported in 36.5% of *E. faecium* strains isolated from blood samples in Australia [10]. In addition, Sievert et al. reported that vancomycin-resistant was found in 82.6% of *E. faecium* strains from the USA [1]. In our study, all of VREfm had the *vanA* gene. Similarly, the majority of VREfm cases worldwide are due to *vanA* with some exceptions to *vanB* in Australia [5, 7, 10, 11, 20]. The *vanA* gene confers high resistance to both vancomycin and teicoplanin. We noted an increase in the consumption of vancomycin and teicoplanin in our hospital. VREfm not only leaves fewer options for disease management but is also important due to the potential risk of vancomycin resistance gene transfer from enterococci to *Staphylococcus aureus*, as previously reported decreased susceptibility to vancomycin in MRSA from our hospital [25].

We analyzed hematological malignancies and previous hospitalization as independent predictor for bacteremia due to VREfm. Both factors facilitate and increase the migration of intestinal microorganisms, such as VREfm, into the bloodstream. The frequent use of chemotherapy in hematological malignancies leads to immunosuppression, which is associated with malnutrition, followed by an inadequate immune response to bacteremia [26]. The presence of frequent comorbid conditions, such as hematological malignancies, prolonged duration of hospitalization, and increased severity of chronic and acute illness (higher Charlson comorbidity index and McCabe score), resulted in high mortality. The increased severity of chronic and acute illness increases the risk of sepsis, prolonged duration of

ICU stay, and high resistance to antibiotics, such as vancomycin and ampicillin in *E. faecium* [2, 10]. Global enterococcal infections, mainly bacteremia caused by MDR *E. faecium*, have become a major problem [11, 27, 28]. Similarly, the levels of resistance to multiple antibiotics and MDR were consistently high in our study. The emergence of MDR VREfm may lead to incurable infections. All of our isolates of *E. faecium* were susceptible to linezolid. Continuous surveillance will be required, however, as recent reports have shown that some VREfm isolates have extremely limited treatment options due to resistance to most useful antibiotics, such as linezolid, tedizolid, ampicillin, daptomycin, and dalbavancin [29].

In this study, VREfm isolates were more resistant to multiple antibiotics and increased MDR activity compared with VSEfm. However, VSEfm strains proliferate in healthcare settings where antimicrobial selection pressures are high; they will continue to acquire additional antimicrobial resistance genetic elements, including the transposon Tn1546 and its variants containing the *vanA* gene, while transposon Tn5281 or its variants harbor *aac(6')-aph(2'')* gene [30].

Infective endocarditis is commonly seen with *Enterococcus* bloodstream infections (BSI). Østergaard and colleagues from Denmark showed that the prevalence of infective endocarditis in patients with positive blood cultures was highest due to *E. faecalis* (16.7%) followed by *S. aureus* (10.1%), *Streptococcus* spp. (7.3%), and the *CoNS* (1.6%). However, the role of *E. faecium* in causing BSI-related infective endocarditis was not seen among the four most common causative organisms. *CoNS* (1.6%) was reported as the fourth most common cause of infective endocarditis [31]. Similarly, we reported the prevalence of infective endocarditis in less than 2% of BSI-related *E. faecium* infections.

Previous studies reported the presence of virulence determinants in strains of *E. faecium* [7, 8, 29]. In our study, 54.9% ($n = 95$) of the clinical isolates from *E. faecium* carried the *esp* gene. A total of 95.8% ($n = 91$) of the *esp*-positive isolates were resistant to multiple antibiotics. Significant correlations between *esp* carriage and antimicrobial resistance of gentamicin and glycopeptides were observed similar to previous studies [7, 8, 29]. Strong correlations between *esp* carriage and antimicrobial resistance occurred due to higher conjugation frequencies observed in isolates that were *esp*-positive compared with those that were *esp*-negative [7]. The frequency of detection of the *hyl* gene in our study was similar to previous reports [7, 8]. The *gelE*, *asa1*, *cylA*, and *ace* genes were rarely detected in *E. faecium*, similar to previous studies [7, 29]. MDR *E. faecium* strains were commonly associated with high mortality and morbidity with increased health-care costs due to the acquisition of antibiotic resistance and virulence determinants [2, 10, 11, 27–30].

PFGE typing showed a considerable diversity of *E. faecium* isolates. In our hospital, *E. faecium* tended to be polyclonal, with more than 50% major clusters and some

Table 5. MLST, patient location, antibiotic resistance, virulence traits, diagnosis, and outcome for selected strains of *E. faecium* clones of *E. faecium* strains isolated from blood (*n* = 29) (GenBank accession numbers: MT746109–MT746129)

Isolate	ST	MLST alleles	CC	PFGE	Patient location	Antibiotic resistance profiles	Antibiotic resistance genes	Virulence traits	Diagnosis	Outcome
E-93	ST17	1-1-1-1-1-1-1	CC17	1	Medical ward	AMP, CIP, ERY, RIF	<i>aac(6)-aph(2'')</i> , <i>vanA</i>	–	ALL, DM, septicemia, fever	Death
E-69	ST17	1-1-1-1-1-1-1	CC17	8	Surgical ward	AMP, CHL, CIP, ERY, RIF	<i>aac(6)-aph(2'')</i> , <i>vanA</i>	–	Medulloblastoma	Death
E-133	ST17	1-1-1-1-1-1-1	CC17	9	ICU	AMP, CIP, ERY, RIF	<i>aac(6)-aph(2'')</i>	<i>hyl</i>	Acute liver failure	Death
E-5	ST17	1-1-1-1-1-1-1	CC17	9	Medical ward	AMP, CIP, ERY, RIF, TET	<i>aac(6)-aph(2'')</i> , <i>vanA</i>	–	AML, sepsis	Discharged
E-25	ST17	1-1-1-1-1-1-1	CC17	9	Medical ward	AMP, CIP, ERY, RIF, TET	<i>aac(6)-aph(2'')</i> , <i>vanA</i>	–	B cell ALL	Discharged
E-141	ST17	1-1-1-1-1-1-1	CC17	9	Medical ward	AMP, CIP, ERY, RIF, TET	<i>aac(6)-aph(2'')</i> , <i>vanA</i>	–	AML	Death
E-2	ST17	1-1-1-1-1-1-1	CC17	13	Medical ward	AMP, CHL, CIP, ERY, RIF, TET	<i>aac(6)-aph(2'')</i> , <i>vanA</i>	<i>esp, hyl</i>	T cell lymphoblastic lymphoma	Discharged
E-33	ST80	9-1-1-1-12-1-1	CC17	1	Medical ward	AMP, CIP, ERY, RIF, TET	<i>aac(6)-aph(2'')</i> , <i>vanA</i>	<i>esp</i>	Erythroderma, sepsis	Discharged
E-128	ST80	9-1-1-1-12-1-1	CC17	1	Medical ward	AMP, CIP, ERY, RIF, TET	<i>aac(6)-aph(2'')</i> , <i>vanA</i>	<i>esp</i>	AML, fever	Discharged
E-130	ST80	9-1-1-1-12-1-1	CC17	1	Medical ward	AMP, CIP, ERY, RIF	<i>aac(6)-aph(2'')</i> , <i>vanA</i>	<i>esp</i>	ALL, septicemia	Discharged
E-11	ST80	9-1-1-1-12-1-1	CC17	1	Medical ward	AMP, CIP, ERY, RIF	<i>aac(6)-aph(2'')</i> , <i>vanA</i>	<i>esp, ace</i>	DM, chronic airways disease	Death
E-51	ST80	9-1-1-1-12-1-1	CC17	1	Medical ward	AMP, CIP, ERY, RIF, TET	<i>aac(6)-aph(2'')</i> , <i>vanA</i>	<i>esp, asaI</i>	Severe anemia	Discharged
E-119	ST80	9-1-1-1-12-1-1	CC17	1	ICU	AMP, CIP, ERY, RIF, TET	<i>vanA</i>	<i>esp</i>	Diffuse SAH, aneurysm	Discharged
E-55	ST80	9-1-1-1-12-1-1	CC17	4	Medical ward	AMP, CIP, ERY, RIF, TET	<i>aac(6)-aph(2'')</i> , <i>vanA</i>	<i>esp, hyl, asaI</i>	AML, sepsis	Discharged
E-4	ST80	9-1-1-1-12-1-1	CC17	6	Medical ward	AMP, CIP, ERY	<i>aac(6)-aph(2'')</i> , <i>vanA</i>	<i>esp, hyl</i>	ALL	Discharged
E-115	ST80	9-1-1-1-12-1-1	CC17	6	Surgical ward	AMP, CIP, ERY, RIF	<i>aac(6)-aph(2'')</i> , <i>vanA</i>	<i>esp, hyl</i>	Carcinoma ovary	Discharged
E-116	ST80	9-1-1-1-12-1-1	CC17	6	Surgical ward	AMP, CIP, ERY, RIF	<i>aac(6)-aph(2'')</i> , <i>vanA</i>	<i>esp, hyl</i>	Aplastic anemia, fever	Discharged
E-120	ST80	9-1-1-1-12-1-1	CC17	6	ICU	AMP, CIP, ERY, RIF	<i>aac(6)-aph(2'')</i> , <i>vanA</i>	<i>esp, hyl</i>	Acute pancreatitis, sepsis	Death
E-114	ST80	9-1-1-1-12-1-1	CC17	7	Medical ward	AMP, CIP, ERY, RIF	<i>aac(6)-aph(2'')</i> , <i>vanA</i>	<i>esp, hyl</i>	Fever, infection, septicemia	Discharged
E-121	ST80	9-1-1-1-12-1-1	CC17	7	ICU	AMP, CIP, ERY, RIF	<i>aac(6)-aph(2'')</i> , <i>vanA</i>	–	COPD	Discharged
E-127	ST80	9-1-1-1-12-1-1	CC17	7	ICU	AMP, CIP, ERY, RIF	<i>aac(6)-aph(2'')</i> , <i>vanA</i>	–	Obstructive inguinal hernia, asthma	Death
E-81	ST80	9-1-1-1-12-1-1	CC17	9	Medical ward	AMP, CIP, ERY, RIF	<i>aac(6)-aph(2'')</i> , <i>vanA</i>	<i>hyl</i>	AML, fever	Death
E-122	ST80	9-1-1-1-12-1-1	CC17	11	Medical ward	AMP, CIP, ERY, RIF	<i>aac(6)-aph(2'')</i> , <i>vanA</i>	–	AML, fever	Discharged
E-112	ST80	9-1-1-1-12-1-1	CC17	16	Surgical ward	AMP, CIP, ERY, RIF, TET	<i>aac(6)-aph(2'')</i> , <i>vanA</i>	–	Cranial tumor	Death
E-124	ST80	9-1-1-1-12-1-1	CC17	18	ICU	AMP, CIP, ERY, RIF, TET	<i>aac(6)-aph(2'')</i> , <i>vanA</i>	<i>esp, hyl</i>	DM, diabetic nephropathy	Discharged
E-34	ST80	9-1-1-1-12-1-1	CC17	18	Surgical ward	AMP, CIP, ERY, RIF, TET	<i>aac(6)-aph(2'')</i> , <i>vanA</i>	<i>esp</i>	Fothergill's repair, bleeding	Discharged
E-22	ST117	9-1-1-1-1-1-1	CC17	1	ICU	AMP, CIP, ERY, RIF, TET	<i>aac(6)-aph(2'')</i> , <i>vanA</i>	<i>esp, hyl</i>	Esophageal aspirates	Discharged
E-3	ST132	7-1-1-1-12-1-1	CC17	8	Medical ward	AMP, CIP, ERY	<i>aac(6)-aph(2'')</i> , <i>vanA</i>	<i>esp, hyl, ace</i>	Aplastic anemia, septicemia	Death
E-1	ST280	1-3-1-1-1-1-1	CC17	16	Medical ward	AMP, CHL, CIP, ERY, RIF, TET	<i>vanA</i>	–	AML	Discharged

MLST, multi-locus sequence typing; ST, sequence type; CC, clonal complex; PFGE, pulsed-field gel electrophoresis; AMP, ampicillin; CHL, chloramphenicol; CIP, ciprofloxacin; ERY, erythromycin; RIF, rifampicin; TET, tetracycline; *aac(6)-aph(2'')*, PCR results for *aac(6)-aph(2'')* genes; *vanA*, PCR results for *vanA* gene; *esp, hyl, asaI*, and *ace*, PCR results for *esp* or *hyl* or *asaI* or *ace* genes; COPD, chronic obstructive pulmonary disease; AML, acute myeloid leukemia; DM, diabetes mellitus; SAH, subarachnoid hemorrhage

minor clones, indicating a diverse population of hospital-acquired *E. faecium* strains due to the exchanges of mobile resistance determinants between *E. faecium* strains. Other studies reported mixed genetic diversity among enterococci isolates originating from different hospitals [20, 27].

Similar to the worldwide report on CC17, all STs belonged to CC17, a highly successful and high-risk clone for the dissemination of antibiotic resistance [11, 28]. Similar to our report, ST80 and/or ST17 of CC17 *E. faecium* strains causing bacteremia have been reported previously in hospitalized patients [10, 11, 28]. ST17 *E. faecium* strain, isolated from the blood sample, first appeared in 1990 in England, quickly became dominant, and has now spread throughout the world [10, 11, 28]. The ST80 VREfm strain isolated from a blood sample of a hospitalized patient first appeared in Israel in 1997. A study by Hammerum and colleagues in Denmark from 2005 to 2015 showed that the prevalence of *vanA*-positive ST80 *E. faecium* strain increased significantly and became one of the dominant STs after ST203 VREfm by replacing many other STs [5]. Similarly, from our MLST results, it appears that the ST80 is dominant in our hospital and may replace the ST17. CC17 isolates expressing *esp* and *hyl* are more likely to bind to fibronectin and laminin than other CCs, which may explain their increased virulence [7, 10, 11, 29].

There were some limitations to this study. All observations were taken from a single hospital and therefore may only reflect local trends and not be representative of other settings in India or other countries. In addition, only representative VREfm strains of the major PFGE clones were genotyped by MLST typing.

In summary, the high proportions of *E. faecium* strains that carry *vanA* and *esp* genes commonly causing bacteremia confirms the extensive infiltration of virulent VREfm genotypes in our hospital. We assume that VSEfm strains proliferate in healthcare settings where antimicrobial selection pressures are high and additional antimicrobial resistance genetic elements will continue to be acquired. The level of antibiotic resistance as well as the accumulation of virulence determinants among *E. faecium* clinical isolates from this study indicated that high-risk CC17 (ST80 and ST17) *E. faecium* had occurred in our hospital. Persistent use of antimicrobial agents in hospitals, in the community, and animal husbandry will certainly further promote the selection and spread of enterococcal MDR clones. Continued surveillance and improved infection control procedures in hospitals are therefore essential to curb the further spread of these high-risk clones.

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Compliance with ethical standards

Conflict of interest The authors declare that they have no conflict of interest.

Ethical approval and informed consent This study was approved by the Ethics Committee of the AIIMS, New Delhi, India (Ref. No.: IEST/T-151/2010). The informed consent of each patient included in this study was obtained.

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